

v1.6.0

- News features:
 - Attachment Connection for FreeBuilding elements
 - Structural Stability feature for AdvancedGridBuilding
 - New position validations
 - Build On Top Of Element
 - Grounded Check
 - Manager can be started in only Destroy mode
 - Improved rotation alignment to the BuildingElements for the FreeBuilding
 - Validation Collider size modifier
- Changes / Refactors:
 - Menu bar Tools refactor
 - Introduced new checks for the Scene Helper Analyze feature
 - New guid based find options for the BuildingElements on the Building
 - Change in the API functions
 - History feature refactor because of the bugs and for maintainability
 - Failure Handling for Should Snap to Foundation
 - CanSnapUnderground property has been removed
 - Gizmos settings are repositioned into the ProjectSettings
 - RaycastSettings ScriptableObject has been removed
 - AreaSettings ScriptableObject has been removed
 - From now the ForcedFallback can not be used for not foundation elements if the UseFoundation logic enabled
 - In the case of the AdvancedGrid algorithm the elements can be only overridden if both the target and the new element have the same snapping ruleset.
 - Blocking of the BuildingManager's input can be done when the manager is inactive
 - BuildingElementList new checks
- Bug Fixes:
 - Fixed multiple issues in the History feature
 - Fixed an issue where the objects Instance Guid generation doesn't work
 - Fixed an issue where the loaded Buildings lost their Parent connection
 - Fixed an issue where the BuildingParent's did not set the parent reference to their buildings
 - Fixed an issue where the elements could coiled with other elements because the ground alignment ignored the BuildingElements
 - Fixed an issue where the AdvancedGridBuilding Ruleset's failure handling for the denied positions did not worked
 - Fixed an issue where the AdvancedGridBuilding Ruleset denied position caching did not worked
 - Fixed an issue where the BuildingManager did not recognised that if the tracker object is null
 - Fixed an issue where the AdvancedGridBuilding algorithm's snapping does not work in some edgcase

- Fixed an issue where the Area could not block the building of the BuildingElements
- Fixed an issue where the Drawer Instance cause stackoverflow
- Fixed an issue where the BuildingManager do not cancel the drag destroy feature in case of deactivation
- Fixed an issue when the placing of an element did exceeded the maximum limit of a building caused crash
- Fixed an issue when the BuildingParent persistent load feature duplicated the default buildings
- Fixed an issue when the AdvancedGrid rules did not applied from the neighbour point of view
- Fixed an issue when the world position has been used instead of the local position and the Buildings are created on a wrong position
- Fixed an issue where the ShouldSnapToFoundation requirement did not checked if the element had no snapping neighbour
- Fixed an issue where the The Snapping does not work if the position of the objects are not exactly in the center of the world
- Fixed an issue where the Element Collision validation did not checked the BuildingElementLinks
- Fixed an issue where the Element Collision validation did not worked because the validation result had been lost

v1.5.0

- News features:
 - Improved Override Element Ruleset
 - Improved Persistency System (Save and Load)
 - New API error codes
 - New and Updated Tools
 - AdvancedGrid SnapPoint RuleSet
 - BuildingElement Link object
 - Buildable Ground Validation
 - BuildingManager's InputInterface can be blocked (or unblocked) temporary
- Changes / Refactors:
 - Every class, enum etc got the "ABS_" prefix.
 - Namespace and folder structure changes
 - Gizmo changes
 - Cache improvements for better performance
 - AdvancedGridBuilding support for "Allow building in the air" feature
 - The AdvancedGridBuilding has a new AdvancedGridType based find and replace option.
 - BuildingElement Metadata removed
 - UI changes
 - Guid settings for the Buildings and the BuildingParents.
- Bug Fixes:
 - Fixed an issue where BuildingElementCreator failed to correctly update the layer of child objects.
 - Fixed an issue where the Highlight feature failed to change all materials in the list
 - Fixed an issue in AdvancedGridBuilding where corner and vertical edge elements could snap during drag building without snapping.
 - Fixed an issue where ForcedFallback caused a disruption in Foundation logic.
 - Fixed an issue where Raycast settings changes were not applied when using the external interfaces.
 - Fixed an issue where grid edge snapping caused visual flashing in both AdvancedGridBuilding and BasicGridBuilding algorithms.
 - Fixed an issue where newly placed BuildingElement instances had faulty collision detection.
 - Fixed an issue where the Foundation Logic did not function correctly.
 - Fixed an issue where the 'Save To File' button on the Building UI incorrectly saved the file path instead of the persisted data.
 - Fixed an issue where BuildingLoader's Parent field incorrectly prevented assigning objects from the scene.

v1.4.1

- Bug Fixes:
 - Fixed an issue where the Demo's camera where black and only the UI can be seen.
 - Fixed an issue where SnappointBased algorithms incorrectly calculated local positions.
 - Fixed an issue where the using VisualScripting broke the build by removing the VisualScripting dependency.
 - Fixed an issue where Floor and Wall elements lacked proper snap points when using AdvancedGridBuilding.

v1.4.0

- News:
 - Object Pool/Buffer
 - Validation changes
 - Collision Based Validation Failure Handling
 - BuildingElement Collision Based Validation
 - Failure Handling
 - It became possible to turn off
 - Building Range Limits
 - New Improved Input System Logic
 - Rotation is able to trigger with button instead of the mouse wheel
 - Every input is able to trigger from message calls
 - New Interface for the message calls
ABS_IBuildingManagerInputActionInterface
 - New input tracker interface ABS_IInputTracker and input tracking functionalities
 - Input keybindings/settings can be changed during runtime.
 - MeshCollision based collision check for SnapPointBased algorithm (Experimental version)
- Changes / Refactors:
 - Tracker
 - renamed function with same functionality:
 - EndOfBuildingProcess > EndOfBuildingCycle
 - NameSpace Changes
 - From the elements are not blocked in case of
 - there are any elements around the temporary element and the snapping is not possible
 - Affected the AdvancedGridBuilding and SnapPointBasedBuilding
 - Clear History
 - From now the ActionHistory can be cleared with the new ClearHistory function on the IBuildingManagerExternalInterface.
- Bug Fixes:
 - AdvancedGridBuilding elements did snap to other elements with different grid sizes.
 - Statistics tool opened the Building Loader instead of the Statistics Window
 - DragBuilding dimension error in case of 1-2 sizes buildings

v1.3.2

- News: -
- Changes / Refactors: -
- Bug Fixes:
 - Build hotfix
 - Rotation fix for AdvancedGrid Building
 - DragBuilding Validation issues with AdvancedGrid Building
 - History Undo/Redo Key Input bug

v1.3.1

- News: -
- Changes / Refactors:
 - Deny the Forced Fallback feature on the BasicGrid and the FreeBuilding algorithms
- Bug Fixes:
 - Rotation bug fixes for AdvancedGridBuilding

v1.3.0

- News:
 - BuildingModes (Single, Continues)
 - Mixed Axes (In case of DragBuilding + AdvancedGridBuilding)
 - Maximum Height for the AirBuilding
 - New Input System support
 - Action History feature (Undo, Redo)
 - New FreeBuilding Rotation Feature
 - Alignment To BuildingElements
 - Maximum Alignment for Ground Alignment
 - InstanceGuid/PrefabGuid
 - BuildingParent
 - BuildingElementList Validation
 - Buildings: Maximum BuildingElement Count
 - Maximum Destroy Count
 - Indestructible elements
 - Prioritize PreBuilt
 - Forced Fallback
- Changes / Refactors
 - Multi-object feature refactor for the BuildingElements.
 - AlignToGround changes
 - Fallback option is removed. In case of fallback the algorithm fallback to freebuilding by default. Based on these changes the fallback freebuilding uses the main algorithm's settings. So the Main algorithm's settings are aligned to fulfill this criteria.
 - Changed Interfaces:
 - IBuildingElementExternalInterface
 - IBuildingManagerExternalInterface
 - IBuildingExternalInterface
 - BuildingManagerTracker
 - The Building's caching is automatically changeable from the editor.
 - The DragBuildingBehaviour property from now available on the BuildingElementMetaData for being able to set it to the type of the elements separately.
- Bug Fixes:
 - Collision Check wrong functionality
 - Multi-Object New Guid generation
 - Should Override Drag Building
 - SnapToGrid BuildingElement Editor feature bug fixes
 - SnapToPreBuiltFinalElement does not work during dragbuilding
 - Empty BuildingManager Crashes
 - UI Editor script bugfixes

v1.2.0

- New Tool: *Statistics*
- New interfaces:
 - *IBuildingManagerStatisticsTracker*
 - *IBuildingExternalInterface*
 - *IBuildingElementExternalInterface*
- New Scriptable Object: LayerCollection
- Custom grid Dimension. From now every axis can have different sizes.
 - BasicGrid : from float to Vector2
 - AdvancedGrid : from float to Vector3
- New functionality for Reposition : Allow Placement During Movement
- New rotation feature: Align Rotation To Ground (Only the FreeBuilding supporting)
- New DragBuilding behavior : Half Frame
- Multi-Object editing support for the following components:
 - Building
 - BuildingElement
 - BuildingElementMetaData
 - BuildingElementHighlightCollection
- New Building GUI features
- From now the guid of the BuildingElement can be changed by hand
- Create button added to the scriptable objects on the GUI
- Info messages added the properties of the scripts
- From now on trackers (including the new statistics trackers) can be added or removed to the BuildingManager during runtime using the new function of the *IBuildingManagerExternalInterface*.
- *IBuildingManagerExternalInterface* got some new functions regardless of the new tracker add and removed functions. The new functions target to get more information about the BuildingManagers state like current Mode, Active BuildingElement etc.
- Increase of the documentation quality
- Refactors for increase the quality of the code
- New HRDP Demo scene for HRDP support
- Bug Fixes
 - Small Demo scene bug fixes
 - Override Element bug fixes
 - Prebuilt Element bug fixes

v1.1.0

- New features
 - Final Element
- Changes
 - Tracker function renaming :
 - FirstElementBlockedStateChanged -> FirstTMPElementBlockedStateChanged
- Fixes
 - The algorithm crashed when an element was in the scene without a valid parent.
 - Added additional checks to the OverrideElement feature to avoid crashing when the feature is enabled but the given object was missing(null).
 - Small bugfixes for the Demo scenes.
 - Removed leftover debug log message in the code.

First release (v1.0.1)

The AdvancedBuildingSystem is starting with the following features:

- Main features
 - Simple Building
 - Drag Building
 - Simple Destroy
 - Drag Destroy
- Different building algorithms
 - Free Building
 - Basic Grid Building (2D grid snapping in 3D space)
 - Advanced Grid Building (3D grid snapping)
 - SnapPointBased (**Experimental!**)
- Customizable settings
 - Position, Rotation, Raycast, Layers, Sizes etc
- Custom Keybinding
- Prebuilt Element feature
- Area features
- Foundation logic
- BuildingElement modifications
 - Override already built element
- Blocking elements feature
- Highlight material feature
- Tracking / API (Custom event system)
- Gizmos
- Persistency (Save/Load)
- Debug (Currently only performance measurements)
- Support
 - NavMesh
- Developer Tools